28.05.26 (M)

Sr. No. of Question paper: 13224

Roll No. ....

Unique paper Code : 32231403
Name of the Course : B.Sc. (H) Zoology
Name of the paper : Biochemistry of Metabolic Processes
Semester : IV
Maximum Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll number on the top immediately after the receipt of the Question Paper.
2. Attempt **five** questions in all.
3. **Question No. 1 is compulsory.**

Q1. a) Define the following terms: (1X7=7)

a. Shuttle system
b. Covalent modification
c. ATP synthase
d. Cytochromes
e. Turnover number
f. Acidosis
g. Coupled reactions

B) Differentiate between the following pairs of terms: (5X2=10)

a. Catabolism and hexokinase
b. Glucogenic and Ketogenic amino acids
c. Synthase and Synthetase
d. Transferase and Isomerase
e. Ligase & Lyase

C) Expand the following terms: (1X5=5)

i. PEP
ii. HMP
iii. AST
iv. PFK
v. SDH

D) Name the cofactor/coenzyme required for the following enzymes: (1X5=5)

i. Pyruvate dhydrogenases
ii. Hexokinase
iii. Citrate synthase
iv. Pyruvates kinase
v. Cytochrome oxidase

2. (a) With the help of structural formulae, describe the fate of glucose under aerobic condition. (8)

   (b) Add a note on ketone bodies. (4)

3. (a) Explain the chemiosmotic hypothesis. (6)

   (b) Elucidate the metabolic pathway of biosynthesis of Palmitic acid. (6)

4. Describe the process of biosynthesis of urea with the help of structures. (12)

5. (a) Describe tricarboxylic acid cycle. (8)

   (b) What is significance of pentose phosphate pathway? (4)

6. (a) Explain the Q cycle and its importance in the Electron transport chaing. (4)

   (b) Give a detail account of biosynthesis of palmitic acid. (8)

7. Write short notes on any **three**: (4x3=12)

   a. Cori Cycleis
   b. Ketogenesis
   c. Malate-aspartate shuttle
   d. Compartmentalization of metabolic pathways in a cell